SE Lab

Lab 1: Requirements Engineering & UML Use-Case Modelling

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SECTION:5B

11.Telemedicine Slot Booking & Prescription Portal

Deliverables

1. Requirements Table

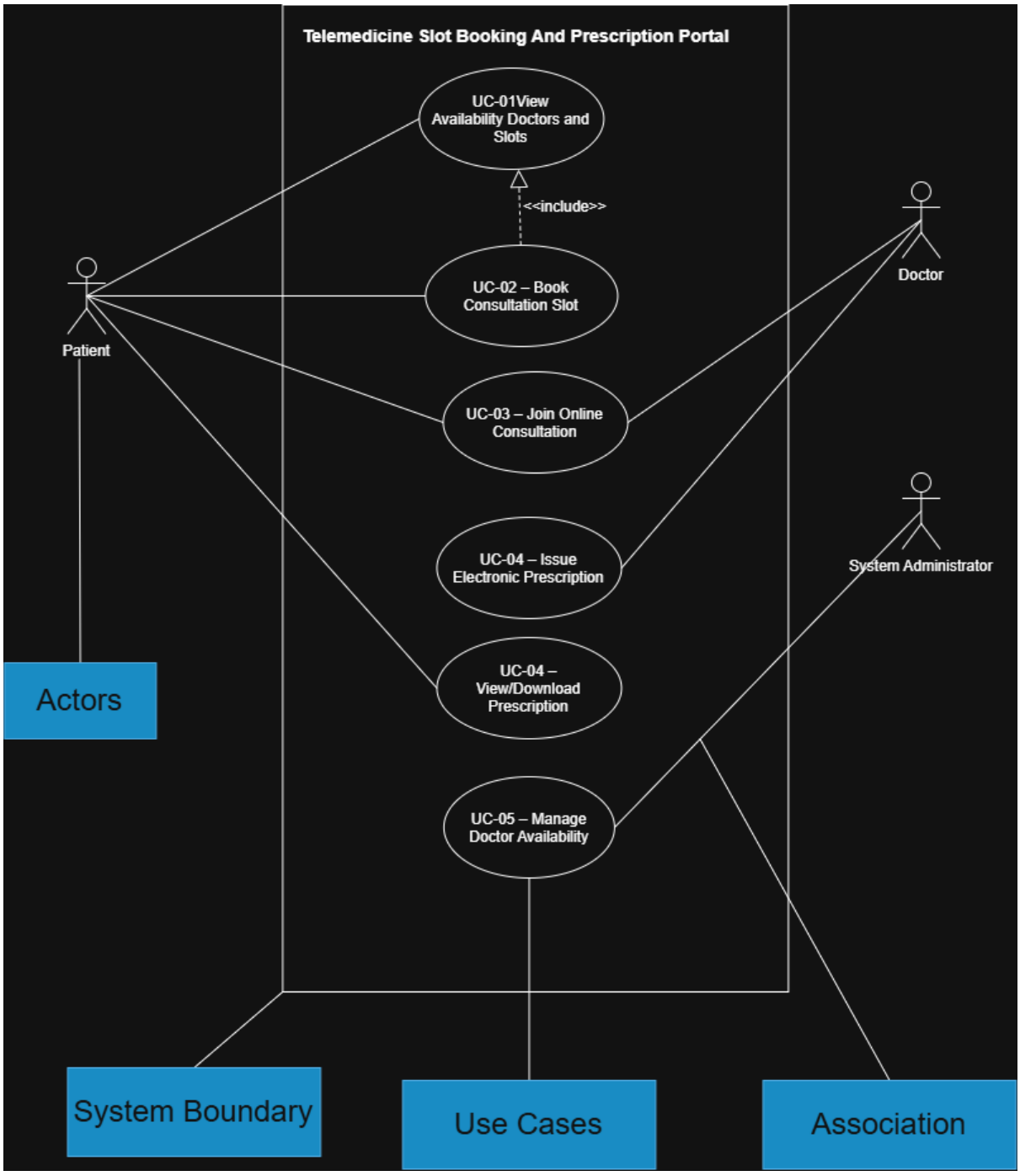
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (Short)
FR-001	Functional	The system shall allow a patient to view available doctors and their consultation slots based on selected specialty and date.	High	When a patient selects a specialty and date, the system shall display only doctors with available slots for that particular date and speciality selection.	Enables patients to find suitable doctors and available appointment times.

FR-002	Functional	The system shall allow a patient to book an available consultation slot with a selected doctor.	High	When a patient selects an available slot and confirms the booking, the system shall create an appointment and display a unique appointment ID.	Appointment booking is the core function of the portal.
FR-003	Functional	The system shall allow a patient to join a scheduled online consultation from the booked appointment.	High	For a valid booked appointment within its scheduled consultation time, the system shall provide an option to join the consultation. The unauthorized patients shall not be allowed to join.	Enables the actual telemedicine consultation while preventing unauthorized access.
FR-004	Functional	The system shall allow a doctor to create and issue an electronic prescription for a patient after a consultation	High	After a doctor submits a valid prescription containing at least one medicine and dosage for a completed consultation, the system	Gives patients convenient access to their medical prescriptions and gives patients convenient access to

		and allow a patient to view and download prescriptions associated with completed consultation		shall save the prescription and make it available to the corresponding patient. The patient shall be able to view the prescription and download it as a file.	their medical prescriptions
FR-005	Functional	The system shall allow a system administrator to manage doctors' consultation availability and appointment slots.	Medium	When the administrator adds, updates, or removes a doctor's available consultation slot, the system shall save the change and reflect the updated availability when patients view available doctors and slots.	Gives patients convenient access to their medical prescriptions.
NFR-001	Non-Functional	The system shall display available appointment slots within 5 seconds of a patient	High	In a test of 20 consecutive searches, at least 19 searches shall display available slots	Ensures responsive slot booking during normal usage.

		submitting a doctor and date search request.		within 5 seconds.	
NFR-002	Non-Functional	The system shall restrict access to patient prescriptions to the authenticated patient and authorized doctor associated with the consultation.	High	During security testing, an authenticated patient shall be able to access only their own prescriptions, and an unrelated patient shall receive an access-denied response for another patient's prescription.	Protects sensitive medical information from unauthorized access.

2. UML Use-Case Diagram



3. Use-Case Flow Document

Use Case: UC-02 – Book Consultation Slot
Primary Actor: Patient
Supporting Actor: System Administrator (because availability is managed by the administrator)

Main Success Scenario

1. Patient logs into the Telemedicine Slot Booking & Prescription Portal.
2. Patient selects the required medical specialty and consultation date.
3. System displays available doctors and consultation slots.
4. Patient selects a doctor and an available consultation slot based on their needs and convenience.
5. System validates that the selected slot is still available.
6. System reserves the selected slot for the patient.
7. System generates a unique appointment ID.
8. System displays the booking confirmation to the patient.
9. The booked appointment is added to the patient's upcoming appointments.
10. Use case ends successfully.

Alternate Flow — Slot No Longer Available

4a. Selected slot becomes unavailable

1. Patient selects a consultation slot.
2. System detects that the slot has already been booked by another patient.
3. System displays a message indicating that the selected slot is no longer available.
4. System returns the patient to the available-slot list.
5. Patient selects another available slot.
6. The use case continues from Step 5 of the Main Success Scenario.